



Emotional Storyteller for Vision Impaired and Hearing-Impaired Children.

(EMOTIONALLY EXPRESSIVE READER FOR CHILDREN WITH SENSORY
CHALLENGES)

EMOTIONAL READER

Project Final Report

J.Brinthapan-IT22123954

A.Sc. (Hons) in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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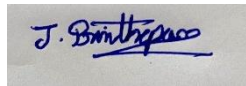
Department of Information Technology

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DECLARATION

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(Ms.Karthiga Rajendran)

ABSTRACT

With the growing emphasis on inclusive education, the need for innovative tools to support children with sensory challenges has become increasingly critical. Deaf and hard-of-hearing children often face barriers in accessing written content, particularly in understanding emotional contexts embedded in text. This project aims to address these challenges by developing an Emotionally Expressive Reader that translates written text into sign language, enriched with emotional expressions. The primary goal is to create an interactive, engaging, and accessible system that not only converts text into sign language but also conveys the emotional tone, thereby enhancing comprehension and storytelling experiences for children. The core of this study involves designing a text-to-sign translation engine integrated with natural language processing (NLP), sentiment analysis, and an avatar-based sign language interpreter. NLP ensures accurate mapping of text into sign language glosses, while sentiment analysis identifies emotional cues and integrates them into the avatar's facial expressions and gestures. The system will be deployed through a child-friendly mobile/tablet application, providing real-time translation, replay options, and interactive features to improve accessibility. This research will investigate the existing challenges in text-to-sign translation, such as limited expressiveness, lack of emotional context, and usability gaps in current systems. By leveraging AI-driven sentiment analysis and avatar-based animation, the proposed solution will address these challenges, contributing to more inclusive and engaging digital learning tools for children with sensory needs. This study forms part of a broader initiative to enhance accessibility and inclusivity in educational technologies.

Keywords: Text-to-Sign Language, Deaf Education, Emotionally Expressive Reader, Natural Language Processing, Sentiment Analysis, Sign Language Animation, Inclusive Learning

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
SL	Sign Language
ESL	Emotionally Expressive Sign Language
OpenGL	Open Graphics Library (used for 3D animation rendering)
FPS	Frames Per Second (animation smoothness)
GPU	Graphics Processing Unit
CNN	Convolutional Neural Network (for gesture/emotion recognition if used)
RNN	Recurrent Neural Network (for sequence/text processing)
LSTM	Long Short-Term Memory (for handling text sequences and emotions)
TTS	Text-to-Speech (optional narration support)
HCI	Human-Computer Interaction
SUS	System Usability Scale (for evaluation)
3D	Three-Dimensional (avatar animation space)
API	Application Programming Interface

I. INTRODUCTION

Children with sensory challenges, particularly those who are deaf or hard of hearing, often encounter significant barriers in accessing educational resources in ways that are both meaningful and engaging. Traditional reading tools rely heavily on text or audio, which may not fully address the communication needs of these learners. More importantly, such tools often lack the ability to convey the emotional context of content—an essential element for comprehension, empathy, and expressive language development. While some digital solutions, such as text-to-speech or basic sign language animation tools, have been introduced, they typically focus only on functional delivery of content and overlook the expressive and emotional dimension of communication. This limitation can lead to reduced engagement, diminished comprehension, and weaker emotional connection to learning materials. Recent advances in artificial intelligence, natural language processing, and animated avatar technology have opened new opportunities to create more inclusive and emotionally intelligent educational tools. By integrating emotion recognition and expressive sign language animation, it becomes possible to design a system that not only translates text into sign language but also embeds facial expressions, gestures, and emotional tone, thereby making communication more natural and impactful. This research proposes the development of an Emotionally Expressive Reader that converts written text into expressive sign language animations for children with sensory challenges. Unlike earlier systems that emphasize basic accessibility, this project focuses on enhancing emotional engagement, comprehension, and inclusivity in education. The proposed solution aims to support children in better understanding learning materials, while also fostering stronger connections between content, emotion, and communication. In doing so, this project contributes to the broader vision of inclusive education technologies that empower children with sensory challenges to learn, grow, and thrive in an environment that respects both their linguistic and emotional needs.

1.1 Background Study and Literature Review

1.1.1 Background Study

Inclusive education is one of the most critical challenges in today's society, especially for children with sensory difficulties such as hearing impairments. Traditional educational methods rely heavily on text and spoken language, making it difficult for children with sensory challenges to fully grasp learning content. While sign language serves as a primary mode of communication for many deaf and hard-of-hearing learners, existing digital learning platforms often fail to incorporate expressive and emotional elements that are crucial for effective communication.



Figure 1: Sign language translation

Reading is not only about decoding text but also about understanding the emotional tone, intent, and context behind it. However, most existing text-to-sign language tools lack emotional depth, focusing only on the literal translation of words. This often results in limited comprehension, reduced engagement, and a diminished learning experience. Advancements in artificial intelligence (AI), natural language processing (NLP), computer vision, and 3D animation now provide opportunities to design systems that can integrate emotion recognition and expressive sign language animations. Such systems can help children not only understand words but also feel the emotional meaning behind them, thereby creating a more engaging and supportive learning environment.

This research aims to build upon existing work by developing an Emotionally Expressive Reader that transforms written text into expressive sign language animations with integrated facial cues and emotional tones. Unlike traditional systems that prioritize accessibility alone, this project emphasizes emotional engagement and inclusivity, addressing a major gap in current educational technologies for sensory-challenged learners. Researchers have explored a variety of approaches to support inclusive learning for children with sensory challenges. Several studies

have focused on sign language recognition and translation systems, while others have explored the role of emotionally intelligent avatars and assistive technologies. However, few solutions integrate both accessibility and emotional expressiveness. One study investigated the development of automatic sign language translation systems that convert text or speech into animated sign gestures. While these systems improved accessibility, they were limited in their ability to capture the emotional nuances of communication, which are essential in sign language expression. [1] Another research initiative explored the use of avatar-based sign language interpreters for digital learning platforms.

These avatars were capable of delivering sign gestures but often lacked realistic facial expressions and emotional cues, making the interaction less engaging for children. [2] Studies on emotion recognition in education have demonstrated that learners benefit significantly when emotional context is integrated into digital tools. For example, systems that used sentiment analysis to detect tone in text improved comprehension and engagement among children. However, these systems were not directly applied to sign language translation. [3] Research on assistive technologies for deaf education has also highlighted the importance of combining visual, expressive, and interactive elements. One project developed an interactive storytelling platform for deaf children, using visual animations to enhance comprehension.

While successful in improving learning outcomes, it lacked direct text-to-sign translation and emotional depth. [4] More recent approaches have begun experimenting with AI-driven expressive avatars that use facial expressions and gestures to simulate emotions. These systems demonstrated potential for creating more natural communication experiences but have yet to be applied specifically to text-to-sign translation for educational use. [5] The gap in current literature lies in the integration of text-to-sign language translation with emotional expressiveness, designed specifically for children with sensory challenges. Most existing systems either prioritize accessibility (literal translation) or emotional intelligence (avatars with expressions), but not both together in a unified learning tool. This research seeks to address that gap by developing a system that not only provides sign language translation but also incorporates emotional tones and facial cues, creating a richer and more inclusive educational experience. By doing so, it will advance the field of assistive learning technologies and contribute to the broader vision of inclusive education.

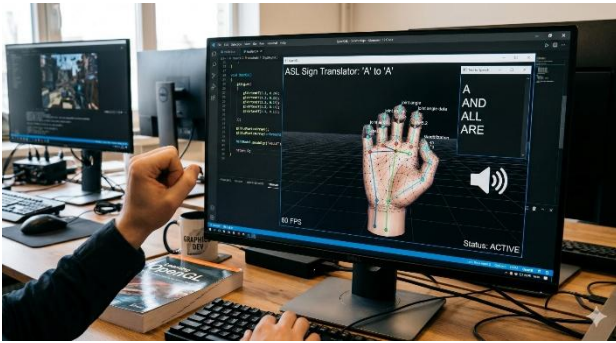


Figure 2 : sign identification

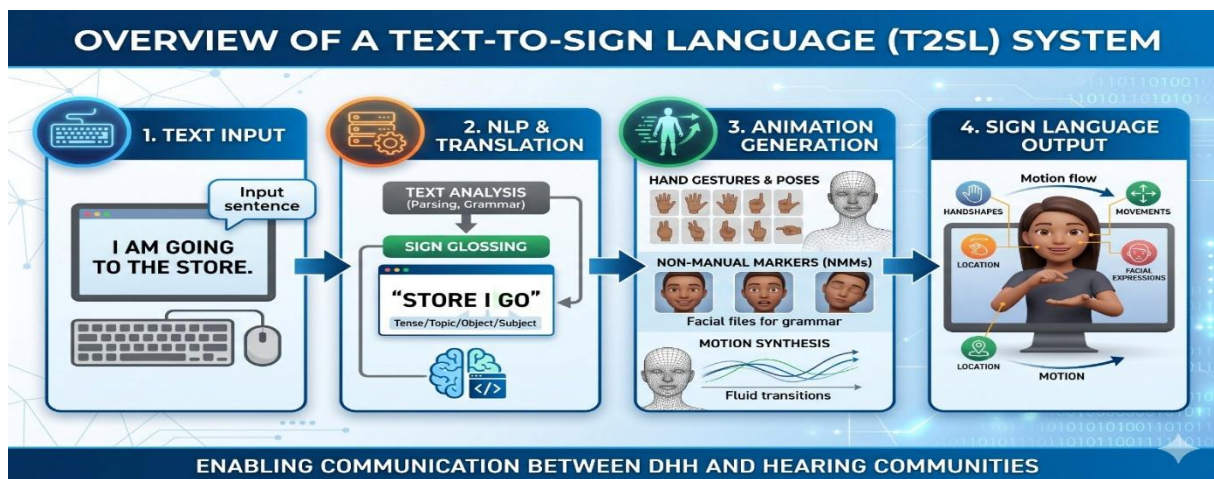


Figure 3 :System Overview

1.2 Literature Review

Inclusive education remains a significant global challenge, particularly for children with sensory impairments such as hearing disabilities. Traditional learning environments rely heavily on spoken language and written text, which limits accessibility for deaf and hard-of-hearing learners. Although sign language serves as a primary communication method for these individuals, many digital learning platforms fail to incorporate expressive and emotional elements that are essential for meaningful communication.

Understanding written content involves more than simple word recognition; it also requires interpreting emotional tone, intent, and context. However, most existing text-to-sign language translation systems focus primarily on literal word conversion, lacking the ability to convey emotional depth. This limitation often leads to reduced comprehension, lower engagement levels, and a less effective learning experience for children. With advancements in Artificial Intelligence (AI), Natural Language Processing (NLP), computer vision, and 3D animation technologies such as OpenGL, there is growing potential to develop systems that integrate emotional awareness into sign language representation.

Several researchers have explored different approaches to support inclusive education for children with sensory challenges. One major area of focus has been automatic sign language translation systems, which convert text or speech into animated gestures. These systems significantly improve accessibility; however, they often fail to capture emotional nuances, which are a critical component of sign language communication [1].

Another line of research involves avatar-based sign language interpreters used in digital learning platforms. While these avatars are capable of performing sign gestures, they frequently lack realistic facial expressions and emotional cues. As a result, the interaction becomes less engaging and less effective for young learners [2].

Studies in emotion-aware educational systems have shown that incorporating emotional context can significantly enhance learning outcomes. For instance, systems utilizing sentiment analysis to detect emotional tone in text have demonstrated improved comprehension and engagement among students. Despite these advancements, such techniques have not been widely integrated into sign language translation systems [3].

Assistive technologies designed for deaf education have also emphasized the importance of visual and interactive learning environments. One study developed an interactive storytelling platform using animations to improve comprehension among deaf children. Although the system successfully enhanced engagement, it lacked direct text-to-sign translation capabilities and did not incorporate emotional expressiveness [4].

More recent developments in AI have introduced expressive virtual avatars capable of simulating human emotions through facial expressions and body movements. These systems show strong potential for creating natural and engaging communication experiences. However, their

application in educational tools—particularly in text-to-sign language translation with emotional context—remains limited [5].

Overall, the existing literature highlights a clear gap: current systems tend to focus either on accessibility through sign language translation or on emotional intelligence through expressive avatars, but rarely integrate both aspects into a single solution. This research aims to address this gap by developing an Emotionally Expressive Reader that combines text-to-sign language translation with emotional cues and facial expressions using OpenGL-based 3D animation. By integrating these components, the proposed system seeks to provide a more engaging, effective, and inclusive learning experience for children with sensory impairments.

1.3 Research Gap

Identifying research gaps is crucial for advancing inclusive education and ensuring that new studies address the unmet needs of children with sensory challenges. While significant progress has been made in assistive technologies, existing research has not fully explored the integration of sign language translation with emotional expressiveness. This leaves a major opportunity for innovation, as emotional cues are vital for improving comprehension, engagement, and overall communication effectiveness. A review of existing literature highlights several areas where this study, the development of an Emotionally Expressive Reader, aims to contribute.

The first research paper [1] discusses automatic text-to-sign language translation systems that convert spoken or written words into sign gestures. While these systems improve accessibility, they are limited to literal translations and often lack emotional depth, which is essential in sign language communication. Emotional cues such as facial expressions and body posture are often excluded, reducing the effectiveness of communication. Therefore, there is a gap in creating systems that combine both linguistic accuracy and emotional expressiveness.

The second research initiative [2] explored avatar-based sign language interpreters for digital learning platforms. Although these avatars enhanced accessibility, they frequently failed to incorporate realistic facial expressions and emotional tones, making the interaction less engaging and less natural. This indicates the need for research that integrates emotion recognition and expressive avatars into text-to-sign systems.

The third research study [3] focused on emotion recognition in educational technologies, showing that learners benefited when digital tools conveyed emotional context. However, these systems primarily targeted text-to-speech applications, not text-to-sign translation. This highlights a gap in applying emotion detection techniques directly to assistive tools for deaf and hard-of-hearing learners.

Another relevant study [4] investigated interactive storytelling platforms for deaf children using visual animations. While the project improved comprehension and engagement, it lacked direct sign language integration and did not address the emotional depth of communication. Therefore,

further research is required to design systems that combine interactive learning, sign language, and emotional cues. Recent studies [5] have experimented with expressive avatars capable of simulating facial emotions. While these show promise for more natural communication, their application in educational settings for sensory-challenged children is still limited. Additionally, few attempts have been made to integrate such avatars into real-time text-to-sign translation tools designed specifically for children. In conclusion, this research identifies a clear gap in existing literature: while studies have addressed either sign language translation or emotional expressiveness, few have integrated both into a single, child-friendly educational tool. The proposed Emotionally Expressive Reader seeks to fill this gap by combining text-to-sign translation, emotion recognition, and expressive avatars, creating a system that not only improves accessibility but also enhances emotional engagement and comprehension. By doing so, this project will advance inclusive education technologies and contribute meaningfully to the development of innovative learning tools for children with sensory challenges

Research Gap Feature	Research 1 [9]	Research 2 [3]	Research 3 [4]	Research 4 [10]	Proposed Research Solution
Integration of Emotional Expressiveness in Sign Language Translation	✓	✗	✗	✗	✓
Real-Time Text-to-Sign Conversion with Emotion Recognition	✗	✗	✗	✗	✓
Child-Centered Assistive Learning Tools Combining Accessibility and Engagement	✗	✓	✗	✓	✓

1.4 Research Problems

The current landscape of assistive learning tools for children with sensory challenges, particularly those who are deaf or hard of hearing, is marked by several limitations that undermine their effectiveness and inclusivity. One of the most pressing problems is the lack of emotional expressiveness in existing text-to-sign language systems. Most available solutions focus solely on providing literal translations of text into sign gestures, but they fail to capture the facial expressions, body language, and emotional tone that are integral components of natural sign language communication. [1] The absence of these features often leads to reduced comprehension, disengagement, and a diminished ability for children to fully grasp the context and meaning of the content.

In addition to this, the majority of current systems rely on static or robotic avatars that lack realism and adaptability. These avatars often deliver signs in a mechanical manner, which does not reflect the dynamic and expressive nature of real human communication. [2] Such limitations make it difficult for children to build a strong emotional connection with the learning material, which is a key factor in sustaining engagement and improving learning outcomes.

Another critical issue lies in the limited accessibility and customization of existing tools. Many assistive systems are designed for general use and do not address the specific needs of children with sensory challenges. They often lack features such as adaptive difficulty levels, culturally relevant sign language variations, or the integration of emotional cues tailored to young learners. [3] This one-size-fits-all approach hinders inclusivity and reduces the overall educational impact.

Furthermore, there is a gap in real-time emotion-aware processing. While some educational technologies have explored sentiment detection for text-to-speech applications, very few have extended this capability to text-to-sign translation with expressive emotional feedback. [4] This lack of realtime emotional adaptation restricts the ability of children to experience natural, interactive, and engaging communication during reading activities.

These challenges highlight the urgent need for a more emotionally expressive, child-centered assistive reader that integrates sign language translation with emotional cues. By combining AI-driven emotion recognition, expressive avatars, and sign language animation, such a system can provide a richer, more engaging, and inclusive learning experience. The proposed Emotionally Expressive Reader aims to fill this gap by enabling children with sensory challenges to not only access information but also connect with its emotional meaning—ultimately supporting better comprehension, engagement, and educational outcomes.

1.5 Research Objectives

Main Objective

The main objective of this research is to develop an Emotionally Expressive Reader that translates written text into sign language with integrated emotional expressiveness (facial expressions, tone, and gestures). The system aims to improve comprehension, engagement, and inclusivity for children with sensory challenges, particularly those who are deaf or hard of hearing

Specific Objectives

- Develop a text-to-sign language translation engine using NLP techniques to convert written text into sign language sequences.
- Train the system using standard sign language datasets (e.g., ASL/ISL/BSL) to improve translation accuracy.
- Validate translation quality through accuracy testing, expert evaluation, and user feedback.
- Implement a sentiment analysis module to detect emotional tone from input text.
- Map detected emotions to facial expressions, body gestures, and signing styles of the avatar.
- Evaluate emotional clarity through user validation with children and educators.
- Design and develop a 3D expressive avatar (using OpenGL) capable of realistic sign gestures and emotional expressions.
- Ensure proper gesture synchronization between hand movements, facial expressions, and text.
- Conduct prototype testing to measure animation quality and user understanding.
- Build an interactive reading platform integrating translation and animation components.
- Implement accessibility features such as playback control, subtitles, and repeat options.
- Perform usability testing to evaluate system performance, engagement, and comprehension.

Business Objectives

- Enhance inclusive education by providing an accessible learning tool for children with hearing impairments.
- Improve learning engagement and comprehension through emotionally expressive sign language animations.
- Provide a child-friendly digital learning experience that supports both education and communication.
- Bridge the gap between text-based learning and sign language understanding with emotional context.
- Support educators and parents by offering an assistive teaching tool for better interaction

2. METHODOLOGY

2.1 Methodology

This research adopts a **modular and iterative development approach** to design and implement an *Emotionally Expressive Reader* that converts written text into sign language with integrated emotional expressions. The system combines Natural Language Processing (NLP), sentiment analysis, 3D animation (using OpenGL), and application development to ensure accurate, expressive, and user-friendly interaction for children with sensory challenges.

3.1 System Architecture Overview

The system is composed of four main modules:

1. **Text Processing Module (NLP)**
2. **Emotion Detection Module**
3. **Sign Language Animation Module (OpenGL-based Avatar)**
4. **Interactive Application Interface**

The workflow follows this pipeline:

Input Text → NLP Processing → Emotion Detection → Sign Mapping → Avatar Animation → User Display

3.2 Text-to-Sign Language Translation Module

This module is responsible for converting written text into structured sign language sequences.

- **Text Preprocessing:**
Input text is cleaned, tokenized, and segmented into meaningful units (words/phrases).
- **Linguistic Analysis:**
NLP techniques such as part-of-speech tagging and syntactic parsing are applied to understand sentence structure.
- **Gloss Mapping:**
Processed text is mapped into **sign language glosses** using a predefined dataset (e.g., ASL/ISL/BSL). Grammar simplification is applied where necessary to match sign language structure.
- **Output Generation:**
A sequence of sign tokens is generated as input for the animation module.

3.3 Emotion and Sentiment Analysis Module

This module identifies the emotional tone of the input text to enhance expressiveness.

- **Sentiment Detection:**
AI-based sentiment analysis models classify text into emotional categories such as *happy*, *sad*, *angry*, *surprised*, or *neutral*.
- **Emotion Mapping:**
Each detected emotion is mapped to predefined **facial expressions, gesture intensity, and animation styles**.
- **Context Handling:**
The system considers sentence-level context to avoid incorrect emotional interpretation.

3.4 Avatar-Based Sign Language Animation (OpenGL)

This module visualizes the output using an expressive animated avatar.

- **3D Avatar Design:**
A child-friendly 3D avatar is created with a skeletal structure (bones/joints) for realistic movement.
- **Gesture Animation:**
Each sign is represented as a sequence of predefined hand and body movements.
- **Facial Expression Integration:**
Emotional data from the sentiment module is applied to the avatar's facial features (eyes, eyebrows, mouth).
- **Synchronization:**
Hand gestures, facial expressions, and timing are synchronized to ensure natural communication.
- **Rendering:**
OpenGL is used for real-time rendering, ensuring smooth animation with optimal FPS.

3.5 Interactive Application Development

A user-friendly application is developed to deliver the system to end users.

- **User Interface Design:**
A simple and intuitive interface is designed for children, enabling easy text input and interaction.
- **Core Features:**
 - Text input or story selection
 - Real-time sign language animation
 - Replay and pause options
 - Adjustable playback speed
 - Visual aids and subtitles
- **Platform:**
The application is designed for mobile/tablet or desktop environments to ensure accessibility.

3.6 Testing and Evaluation

To ensure effectiveness and usability, the system undergoes multiple evaluation stages:

- **Functional Testing:**
Verifies correctness of text-to-sign translation and animation output.
- **Accuracy Evaluation:**
Experts validate the correctness and fluency of sign language sequences.
- **User Testing:**
Conducted with children and educators to measure:
 - Comprehension
 - Engagement
 - Emotional clarity
- **Usability Assessment:**
The System Usability Scale (SUS) is used to evaluate user satisfaction and ease of use.

3.7 Implementation Approach

The development follows an **iterative prototyping model**:

- Initial prototype with basic translation
- Integration of emotion detection
- Enhancement with expressive avatar
- Final deployment with full application features

This approach ensures continuous improvement based on user feedback.

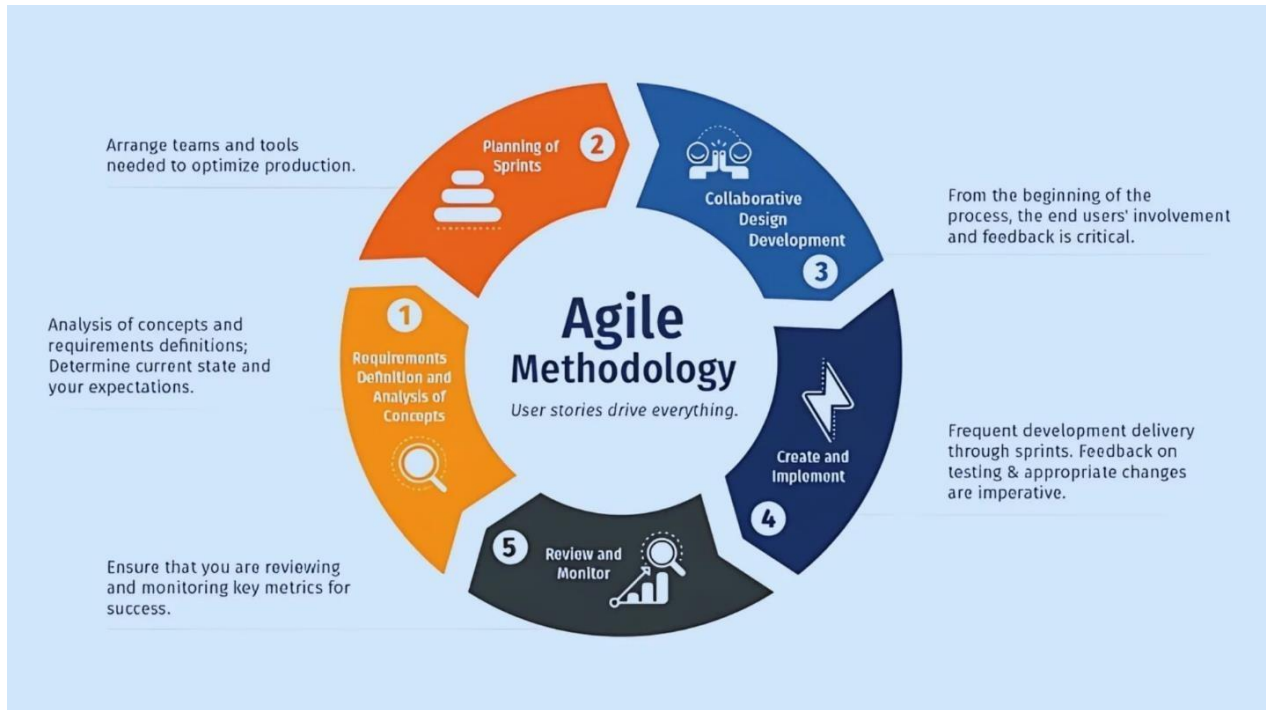


Figure 4 :Agile Development life cycle

2.1.1 Feasibility Study and Planning

Before initiating the development of the *Emotionally Expressive Reader*, a comprehensive feasibility study was conducted to evaluate the practicality and success potential of the proposed system. This analysis focused on technical, economic, operational, and scheduling aspects to ensure a well-structured and achievable implementation.

4.1 Technical Feasibility

Technical feasibility assessed whether the required technologies and tools could effectively support the system's development. The proposed solution integrates Natural Language Processing (NLP), sentiment analysis, and real-time 3D animation using OpenGL to convert text into emotionally expressive sign language.

Modern AI frameworks such as TensorFlow or PyTorch can be utilized for NLP and emotion detection, while OpenGL provides efficient rendering for avatar-based animation. Additionally, existing sign language datasets (e.g., ASL/ISL/BSL) support the training and mapping of sign gestures. These technologies are widely available, well-documented, and capable of handling the system's requirements.

The hardware requirements are moderate, as the system can be optimized to run on standard desktop or mobile devices with GPU support for smooth animation rendering. Therefore, the project is considered technically feasible.

4.2 Economic Feasibility

Economic feasibility evaluated the cost-effectiveness of the project. The system primarily relies on open-source technologies and frameworks, significantly reducing development expenses. Tools for NLP, AI model training, and OpenGL-based rendering are freely available, minimizing licensing costs.

Furthermore, the use of publicly available datasets reduces the need for expensive data collection processes. The system can be developed and deployed with limited financial resources, making it a cost-efficient solution.

In the long term, the project also holds potential for scalability and commercialization as an assistive educational tool, increasing its overall economic value.

4.3 Operational Feasibility

Operational feasibility focused on the usability and practicality of the system in real-world scenarios. The application is designed with a **child-friendly and intuitive interface**, ensuring ease of use for children with hearing impairments as well as educators and parents.

Users will be able to input text or select predefined stories and instantly view expressive sign language animations through an interactive avatar. Features such as replay options, adjustable playback speed, and visual aids enhance accessibility and user engagement.

The system is designed to operate efficiently without requiring advanced technical knowledge, ensuring smooth adoption in educational environments such as schools and learning centers.

4.4 Scheduling Feasibility

Scheduling feasibility examined whether the project could be completed within the allocated timeframe. The development process is structured into clearly defined phases:

- Requirement analysis and system design
- Dataset preparation and preprocessing
- NLP and sentiment analysis model development
- Avatar design and animation (OpenGL integration)
- System integration and application development
- Testing, evaluation, and refinement

An **Agile development approach** is adopted, dividing the project into iterative sprints. This allows continuous progress tracking, flexibility in handling challenges, and incremental improvements based on feedback. With proper planning and task management, the project can be completed within the **مقرر** timeframe.

Gantt Chart

PROCESS	PHASE 1				PHASE 2				PHASE 3			
	AUGUST	SEPTEMBER	OCTOBER	NOVEMBER	DECEMBER	JANUARY	FEBRUARY	MARCH	APRIL	MAY	JUNE	JULY
Feasibility Study												
Environmental Setup												
Proposal Stage												
Development												
Implementation												
Testing												
Final Stage												

Figure 5 :Gantt Chart

Work Breakdown chart

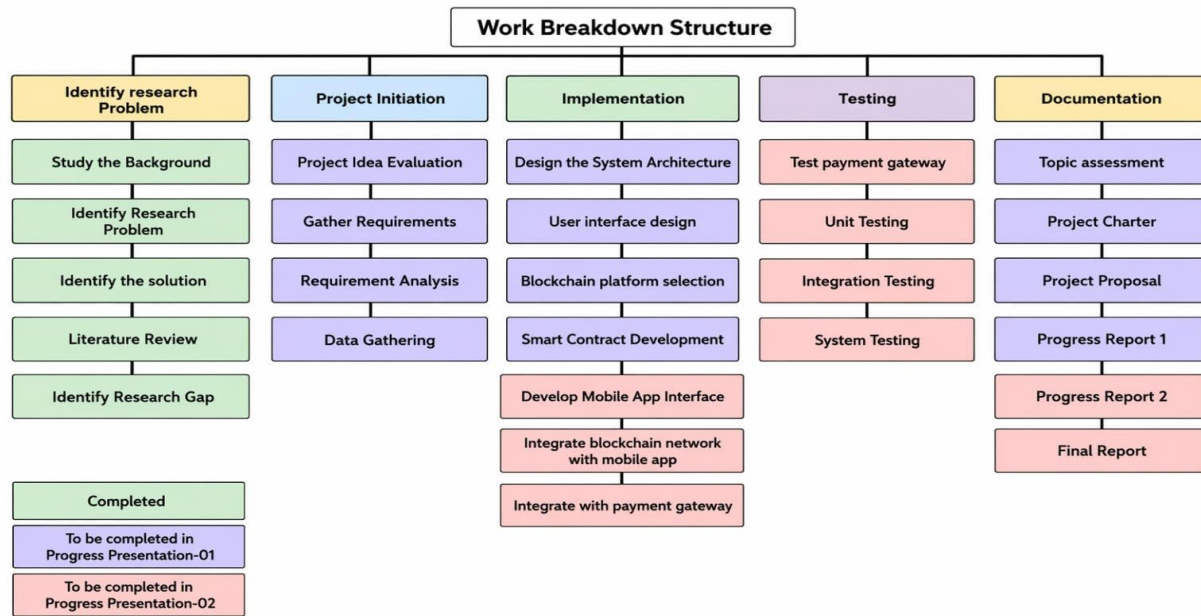


Figure 6 :Work Breakdown chart

2.1.2 Requirement Gathering & Analysis

Functional Requirements:

- Text-to-Sign Language Conversion:

The system must accurately convert input text into corresponding sign language animations. Each word, phrase, or sentence should be translated into standardized sign gestures to ensure effective communication for deaf and hard-of-hearing users.

- Emotionally Expressive Sign Rendering:

The system must integrate emotional expressions (e.g., happy, sad, excited, neutral) into the sign animations. This ensures that the meaning and tone of the message are conveyed effectively, supporting better comprehension for children with sensory challenges.

- Real-Time Translation:

The application must provide real-time text-to-sign translation. As a user types or inputs text, the system should immediately generate and display the corresponding sign language animation without significant delay.

- User-Friendly Interface for Children:

The system must include an intuitive and engaging interface designed for children. This includes features such as clear visuals, simple navigation, and interactive elements to enhance the learning experience.

- Accessibility Features:

The system must include additional accessibility options such as text resizing, customizable color themes, and playback controls for sign animations (pause, replay, speed adjustment) to cater to children with different sensory or learning needs.

- Feedback and Confirmation Mechanism:

After each translation, the system must provide a confirmation output (e.g., visual cues, playback summary, or voice/text feedback) so that users and educators can verify the accuracy of the generated signs

Non-Functional Requirements:

- **Performance:**

The system should process text input and display the corresponding sign language animation within a few seconds to ensure real-time communication and learning support.
- **Scalability:**

The system should be capable of supporting an increasing number of users, multiple devices, and expanding sign language datasets without performance degradation.
- **Security:**

The system must ensure the protection of user data, especially children's information, through encryption and secure authentication mechanisms, preventing unauthorized access or misuse.
- **Reliability:**

The system must consistently deliver accurate text-to-sign translations and maintain high availability, minimizing downtime to ensure uninterrupted learning and communication

User Requirements:

- Ease of Use:

The mobile app should be intuitive and easy to navigate, with a clean and responsive user interface that accommodates users of all technical levels.

- Security and Privacy:

Users expect that their personal and financial information will be protected.

- Accessibility:

The app should be accessible to users with disabilities, incorporating features like voice commands and screen reader compatibility.

System Requirements:

Hardware Requirements:

- Servers or Cloud Infrastructure: To host the text-to-sign translation engine, manage the animation library, and handle real-time processing.
- User Devices: Mobile devices, tablets, or desktop computers for running the application and displaying sign language animations.
- Peripheral Devices (Optional): Cameras and microphones for possible future integration of speech-to-text and gesture recognition features.

Software Requirements:

- Sign Language Translation Engine: Core software module responsible for converting text input into corresponding sign language gestures.
- Animation Rendering Framework: Tools for displaying 3D/2D animated avatars capable of performing expressive sign language movements.
- Database System: To securely store sign language datasets, user preferences, and learning progress data.
- Accessibility Support Libraries: To integrate features such as text resizing, color adjustments, and interactive controls for children with sensory challenges

Use case scenarios

Use Case ID	UC-01
Use Case Name	Text-to-Emotionally Expressive Sign Language Translation
Preconditions	• The user (child/educator) has the application installed and is logged in. • The device is powered on and capable of running the application • The system has access to the required NLP and animation modules..
Primary Actor	Child with hearing impairment
Main Success Scenario	1. The user opens the application. 2. The user enters text or selects a predefined story from the application. 3. The system processes the input text using Natural Language Processing (NLP). 4. The system performs sentiment analysis to detect the emotional tone of the text. 5. The text is converted into corresponding sign language gloss sequences. 6. The OpenGL-based avatar generates synchronized sign language gestures. 7. Facial expressions and emotional cues are applied to the avatar based on detected sentiment. 8. The user views the expressive sign language animation in real time.
Extensions	4a). If the system cannot detect a clear emotion, a neutral expression is applied. 6a). The user can replay, pause, or adjust the speed of the animation. 6b). The user can select different stories or educational content from the library. 6c). Educators or parents can monitor usage and assist children during interaction.

Table 2: Use case from the child

User Stories

User Story 1 – Child User (Learner):

As a child with hearing impairment, I want to enter or select text from the application so that I can understand the content through emotionally expressive sign language animations.

Acceptance Criteria:

- The user can type text or select a story from the app.
- The system converts the text into sign language accurately.
- The avatar displays correct hand gestures synchronized with the text.
- Emotional expressions (e.g., happy, sad, angry) are clearly shown in the animation.
- The user can replay, pause, and control the speed of the animation.

User Story 2 – Educator / Parent:

As an educator or parent, I want to use the application to teach children using expressive sign language so that I can improve their understanding, engagement, and emotional connection to learning content.

Acceptance Criteria:

- The educator can select or input educational content for the child.
- The system presents content using expressive sign language animations.
- Emotional tone is correctly reflected based on the text.
- The interface is simple and child-friendly for guided learning.
- The educator can assist and monitor the child's interaction during usage.

2.1.3 Designing the System

The design of the *Emotionally Expressive Sign Language Translator* focuses on creating an accessible, engaging, and child-friendly system that converts written text into expressive sign language animations. The system is structured using a **modular architecture**, consisting of components for text input, natural language processing, emotion detection, sign language mapping, avatar animation, and user interaction. This modular design ensures flexibility, allowing individual components to be improved or extended without affecting the overall system.

During the design phase, a **user-centered approach** was adopted, with special attention given to children with hearing impairments. The system workflow was simplified to ensure ease of use: the user inputs or selects text, and the system automatically processes it to generate sign language animations with appropriate emotional expressions. The application interface serves as the main interaction point, while the backend integrates NLP models, sentiment analysis, and animation control mechanisms. OpenGL is utilized for rendering the animated avatar, ensuring smooth and real-time visual performance.

The system design prioritizes **accuracy, expressiveness, and usability**. The text processing module analyzes and structures input text to ensure correct linguistic interpretation. The sign language translation module converts processed text into corresponding sign sequences using a predefined dataset. The emotion detection module applies sentiment analysis techniques to identify the emotional tone of the text, which is then mapped to facial expressions and gesture styles of the avatar.

The animation module plays a critical role in delivering the output. A 2D avatar is designed with articulated joints to perform realistic hand gestures and synchronized facial expressions. The system ensures proper **gesture synchronization**, aligning hand movements, facial cues, and timing to reflect both linguistic meaning and emotional context accurately. Real-time rendering through OpenGL enables smooth animations, enhancing the overall user experience.

Finally, the interactive application integrates all components into a cohesive platform. Features such as text input, predefined story selection, playback control, and visual aids are incorporated to improve accessibility and engagement. The system is designed to operate efficiently on standard devices, ensuring wide usability.

Overall, the design aims to deliver a **robust, efficient, and inclusive solution** that combines AI, NLP, and animation technologies to enhance communication and learning for children with hearing impairments. By integrating emotional expressiveness into sign language translation, the system provides a richer and more meaningful educational experience.

2.1.4 System Architecture Diagram

The system architecture of the *Emotionally Expressive Sign Language Translator* illustrates how different components interact to deliver an intuitive and engaging learning experience. The architecture follows a **layered and modular approach**, integrating text processing, emotion detection, sign language translation, 3D animation, and application interface components to generate expressive sign language output from written text.

At the **user interface layer**, children, educators, or parents interact with the system through a mobile or desktop application. This layer allows users to input text manually or select predefined stories and educational content. The interface is designed to be simple, visually appealing, and child-friendly to ensure ease of use.

Once the input is provided, the data is passed to the **text processing layer**, which utilizes Natural Language Processing (NLP) techniques. In this layer, the text is cleaned, tokenized, and analyzed to extract linguistic structure and meaning. This ensures that the input is accurately interpreted before being converted into sign language sequences.

The processed text is then forwarded to the **emotion detection layer**, where sentiment analysis models are applied to identify the emotional tone of the text, such as happiness, sadness, anger, or neutrality. This layer plays a crucial role in enhancing the expressiveness of the system by associating emotional context with the content.

Next, the system moves to the **sign language translation layer**, where the processed text is mapped into corresponding sign language glosses using a predefined dataset (e.g., ASL/ISL/BSL). This layer ensures that the grammatical structure is adapted appropriately for sign language representation.

The **animation and rendering layer** is responsible for visual output. A 2D/3D avatar, developed using OpenGL, performs the sign language gestures. The avatar incorporates both hand movements and facial expressions, which are synchronized with the detected emotional tone. This layer ensures real-time rendering with smooth transitions, maintaining a high frame rate for a natural viewing experience.

Finally, the **application output layer** presents the animated result to the user. The system includes interactive features such as replay, pause, and speed adjustment to improve accessibility and engagement. Optionally, user interactions and learning data can be stored for future improvements and analysis.

The modular structure of the architecture ensures scalability and flexibility, allowing additional features such as multi-language support, advanced emotion detection, or improved animation models to be integrated without affecting existing components.

Figure X presents the system architecture diagram, illustrating the flow of data from user input through NLP processing, emotion detection, sign mapping, and avatar animation to the final output. Arrows indicate the sequence of operations, while each module represents a key functional component. This architecture demonstrates how AI, NLP, and OpenGL-based animation are integrated to create an inclusive and emotionally expressive learning system.

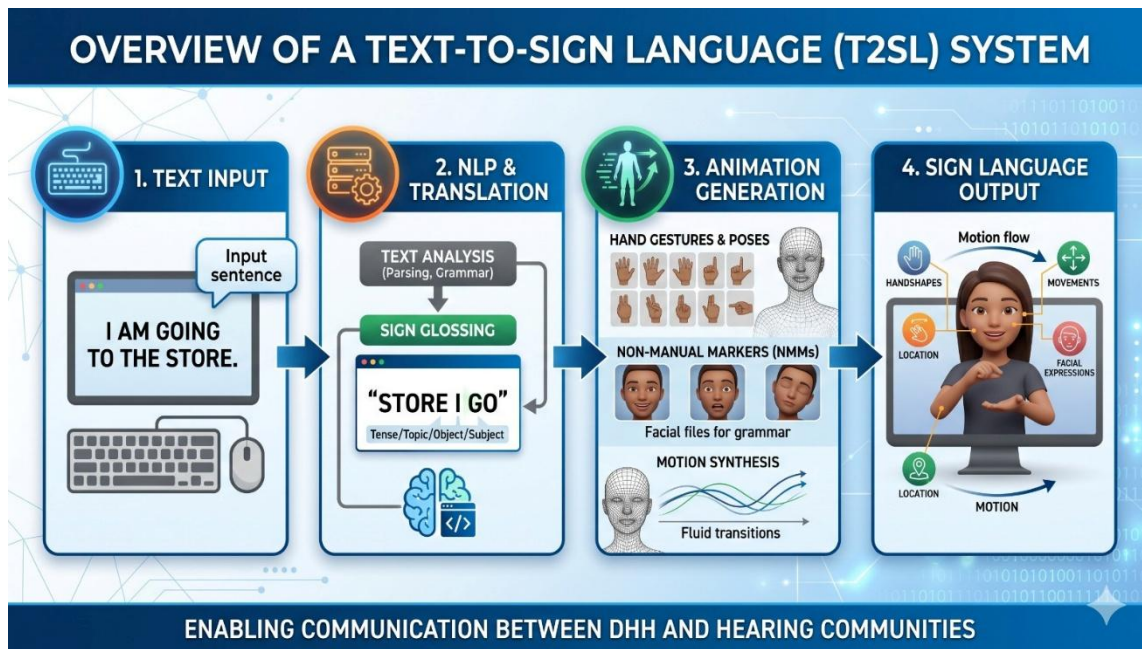


Figure 7 :system architecture diagram

2.1.5 Implementation

The implementation phase focuses on transforming the system design into a functional solution that converts written text into **emotionally expressive sign language animations** through an interactive application. The system was implemented using a combination of technologies, including **Python** for backend processing, **TensorFlow/PyTorch** for NLP and sentiment analysis models, **OpenGL** for real-time avatar animation, and a **mobile/desktop application framework** (e.g., Flutter or React) for the user interface.

The workflow begins at the application layer, where users enter text or select predefined stories. The input is sent to the **text processing module**, where Natural Language Processing (NLP) techniques are applied. This includes tokenization, sentence parsing, and grammatical simplification to prepare the text for sign language translation. The processed text is then converted into structured **sign language gloss sequences** using a predefined dataset.

Next, the text is passed to the **emotion detection module**, where sentiment analysis models classify the emotional tone of the content into categories such as happy, sad, angry, or neutral. This emotional information is crucial for enhancing the expressiveness of the generated animations.

The **sign language translation module** maps the processed text into corresponding sign gestures. These gestures are then forwarded to the **animation module**, where a 2D/3D avatar performs the signs. The avatar is designed with a skeletal structure to enable realistic hand and body movements. Using OpenGL, the system renders animations in real time, ensuring smooth transitions and maintaining a stable frame rate (FPS).

Facial expressions and emotional cues are synchronized with the hand gestures to reflect the detected sentiment accurately. This synchronization ensures that both the **linguistic meaning and emotional context** are conveyed effectively to the user.

The application layer presents the final output, allowing users to view the animated sign language in an interactive environment. Features such as replay, pause, and adjustable playback speed are implemented to improve usability and accessibility for children.

During the implementation phase, **module-level testing** was conducted for each component, including NLP processing, emotion detection, and animation rendering. This was followed by **integration testing** to ensure seamless communication between modules and consistent system performance.

The implementation phase successfully transformed the conceptual design into a working prototype by integrating AI, NLP, and OpenGL-based animation technologies. The resulting system provides a real-time, expressive, and user-friendly solution that enhances communication and learning for children with hearing impairments.

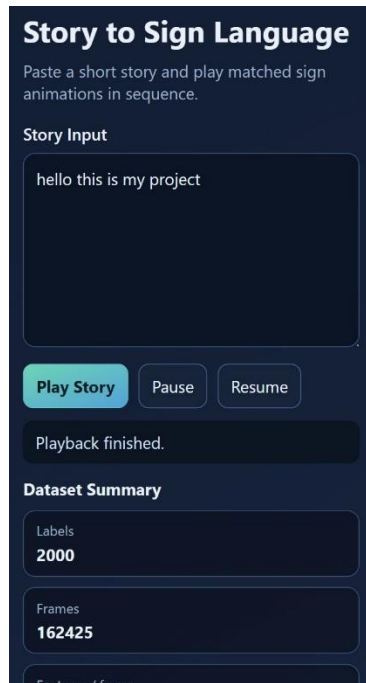


Figure 8 : Mobile interface

```
sign > MyResearch > animation_opengl.py
1 # animation_opengl.py
2 import numpy as np
3 from OpenGL.GL import *
4 from OpenGL.GLU import *
5 import glfw
6 import math
7
8
9 class SignAnimatorOpenGL:
10     def __init__(self, width=900, height=700):
11         self.width = width
12         self.height = height
13
14         self.prev_pose = None
15         self.prev_left = None
16         self.prev_right = None
17         self.smooth_factor = 0.75
18
19         if not glfw.init():
20             raise Exception("GLFW init failed")
21
22         self.window = glfw.create_window(
23             self.width, self.height, "Full Body Sign Avatar (IK)", None, None
24         )
25         glfw.make_context_current(self.window)
26
27         glViewport(0, 0, self.width, self.height)
28         glMatrixMode(GL_PROJECTION)
29         glLoadIdentity()
30         gluOrtho2D(0, self.width, self.height, 0)
31         glMatrixMode(GL_MODELVIEW)
32         glLoadIdentity()
33
34         glClearColor(0.05, 0.05, 0.08, 1)
35
```

Figure 9 : Animation generation

```

jpn > MyResearch > main.py
31 # ===== PLAY ANIMATION =====
32 def play_animation():
33     print("-- Prediction mode")
34     print("="*50)
35
36     # Load model
37     model_wrapper = SignLanguageModel(input_shape=(None, 1), num_classes=1)
38     if not model_wrapper.load(MODEL_FILE):
39         print("Please train the model first!")
40         return
41
42     loader = DataLoader()
43     label_encoder = loader.load_label_encoder()
44     if label_encoder is None:
45         print("Please train the model first!")
46         return
47
48     predictor = SignPredictor(model_wrapper.model, label_encoder)
49     animator = SignAnimatorOpenGL(width=800, height=600)
50
51     print("\nAvailable signs:")
52     available_labels = predictor.list_available_labels()
53     if available_labels:
54         print(f"{'', '.join(available_labels)}")
55
56     print("\nControls while animating:")
57     print("  q or ESC = quit, p = pause/resume, r = restart, n = next sign")
58     print("="*50)
59
60     try:
61         while not animator.should_close():
62             user_input = input("\nEnter sign label(s) or text: ").strip()
63             if user_input.lower() in ["exit", "quit", "q"]:
64                 break

```

Figure 10 : for inference from saved model and output visualization.

```

def train_model():
    print("Training mode")
    print("="*50)

    loader = DataLoader()
    if not loader.load_dataset():
        return

    X_train, X_val, y_train, y_val = loader.prepare_data()
    if X_train is None:
        return

    model = SignLanguageModel(input_shape=(loader.max_len, 1),
                              num_classes=len(loader.label_encoder.classes_))
    model.build_model()
    model.train(X_train, y_train, X_val, y_val)
    model.save(MODEL_FILE)

    if X_val is not None and y_val is not None:
        loss, acc = model.model.evaluate(X_val, y_val, verbose=0)
        print(f"Final validation accuracy: {acc:.2%}")

    print("Training completed!")

```

Figure 11 : train models, training loop

2.1.6 Development Environment and Tools

The development of the Pest Identification and Control System utilized a combination of software and hardware tools selected to ensure accuracy, efficiency, and scalability. The backend of the system was implemented using Python, which provided a flexible platform for integrating deep learning, image processing, and cloud services. The deep learning model, based on VGG16 transfer learning, was developed using TensorFlow and Keras, enabling fast experimentation and reliable model training.

Image preprocessing and leaf severity analysis were performed using OpenCV, which allowed efficient manipulation of images, noise reduction, and segmentation using HSV color space. Data augmentation techniques, such as rotation, flipping, zoom, and brightness adjustments, were applied using TensorFlow's ImageDataGenerator, enhancing the model's ability to generalize under varied field conditions.

For the mobile application, Flutter was chosen to provide a cross-platform solution, allowing the app to run seamlessly on both Android and iOS devices. The app interfaces with the backend through Firebase, which serves as a cloud database for storing detection results, severity levels, and fertilizer recommendations in real-time. Firebase also enables synchronization of historical data, providing farmers and agricultural officers with a convenient way to track pest infestations over time.

The hardware environment for model development and testing included a development PC equipped with a GPU-enabled graphics card to accelerate deep learning training. Mobile testing was performed on smartphones with minimum 2 GB RAM to ensure smooth performance of the Flutter application and real-time inference using TensorFlow Lite.

Summary of tools and libraries:

- Python 3.11 – Backend programming
- TensorFlow / Keras – Deep learning model development
- OpenCV – Image processing and leaf severity analysis
- Flutter – Cross-platform mobile application development
- Firebase – Cloud storage and real-time data synchronization
- GPU-enabled PC – Model training and testing
- TensorFlow Lite – Model deployment on mobile devices

The combination of these tools and technologies provided a robust, scalable, and user-friendly system that integrates AI-based pest identification, leaf severity analysis, and actionable fertilizer recommendations, making it suitable for real-world deployment in paddy farming.

2.1.7 Data Collection

The data collection phase is a critical component in developing the *Emotionally Expressive Sign Language Translator*, as the accuracy of sign generation and animation depends heavily on the quality and consistency of the collected gesture data. Unlike traditional image-based datasets, this system utilizes **real-time webcam capture** to record hand, body, and facial landmark positions, which are then used to build a structured sign language dataset.

The dataset consists of **sign gesture recordings represented as joint coordinate points (landmarks)**. Each sign is captured as a sequence of frames, where key landmarks—such as finger joints, wrist, elbow, and facial points—are detected and recorded. These landmarks are extracted using computer vision frameworks (e.g., MediaPipe or similar libraries), which provide precise tracking of hand and body movements.

Data collection is performed through the following process:

1. **Webcam-Based Capture:**
Users perform predefined sign gestures in front of a webcam. The system captures real-time video and extracts landmark coordinates for each frame.
2. **Landmark Extraction:**
For every captured frame, the system records the positions of key joints (e.g., x, y, z coordinates of fingers, palm, and facial features). This transforms visual gestures into numerical data suitable for model training.
3. **Sequence Recording:**
Each sign is stored as a sequence of frames to preserve motion dynamics, ensuring that temporal information (gesture flow) is retained.
4. **Data Labeling:**
Each recorded sequence is labeled with the corresponding **sign meaning (word/phrase)** and, where applicable, associated **emotional context** (e.g., happy, sad, neutral).
5. **Dataset Storage:**
The collected data is stored in a structured format such as a **CSV file (sign_dataset.csv)**, where each row represents a frame or sequence of landmark coordinates along with its label. This format enables easy preprocessing and integration with machine learning models.

To improve the robustness of the dataset, multiple samples of each sign are recorded under varying conditions, including different users, lighting environments, and gesture speeds. This ensures that the system can generalize well across real-world scenarios. The dataset is then organized into **training and testing subsets**, maintaining a balanced representation of all sign classes. Additional metadata, such as user ID, recording conditions, and timestamp, may also be included to support future enhancements and analysis.

Overall, this data collection approach provides a **rich, structured, and scalable dataset** for training gesture recognition and animation systems. By capturing detailed joint movements and storing them in a machine-readable format, the system establishes a strong foundation for accurate sign language translation and expressive avatar animation.

Figure 12 :Csv file

Figure 13 : Csv file 2

2.1.8 Data Preprocessing

Data preprocessing is a critical step in the development of the *Emotionally Expressive Sign Language Translator*, as it ensures that the captured gesture data is clean, consistent, and suitable for training machine learning models. The raw data collected through webcam-based landmark detection often contains noise, missing values, variations in user posture, and differences in recording speed, all of which can affect model performance if not properly handled.

The preprocessing pipeline begins with **data cleaning**, where incomplete or corrupted landmark frames are removed. Missing joint coordinates are either interpolated or discarded to maintain data integrity. This step ensures that only high-quality sequences are used for training.

Next, **normalization** is applied to the landmark coordinates. Since different users may appear at varying distances and positions relative to the camera, the coordinates are scaled and centered (e.g., relative to a reference joint such as the wrist or torso). This makes the model invariant to position and size differences.

The system then performs **sequence alignment and padding**, as sign gestures vary in length. All sequences are standardized to a fixed number of frames by trimming longer sequences or padding shorter ones. This ensures compatibility with sequence-based models such as LSTM.

To improve model robustness, **data augmentation** techniques are applied. These include slight variations in position, scaling, and temporal speed adjustments to simulate different signing styles. Augmentation helps the model generalize better across different users and environments.

Additionally, **feature extraction** is performed by selecting relevant joint points (e.g., fingers, wrists, facial landmarks) and optionally computing derived features such as angles or distances between joints. This reduces dimensionality while preserving important gesture information.

The processed data is then structured and stored in a format suitable for training, such as normalized sequences in a CSV file or tensor format. Finally, the dataset is divided into **training and testing sets**, ensuring balanced representation across all sign classes and emotional categories.

Overall, the preprocessing phase transforms raw landmark data into a standardized and high-quality dataset, providing a strong foundation for accurate gesture recognition and expressive sign language generation.

```

sign > MyResearch > model.py
10 class SignLanguageModel:
16     def build_model(self):
35         return self.model
36
37     def train(self, X_train, y_train, X_val=None, y_val=None):
38
39         if self.model is None:
40             print(" Model not built yet!")
41             return None
42
43         callbacks = []
44         if X_val is not None and y_val is not None:
45             callbacks.append(
46                 EarlyStopping(
47                     monitor='val_loss',
48                     patience=MODEL_CONFIG['patience'],
49                     restore_best_weights=True
50                 )
51             )
52
53         history = self.model.fit(
54             X_train, y_train,
55             epochs=MODEL_CONFIG['epochs'],
56             batch_size=MODEL_CONFIG['batch_size'],
57             validation_data=(X_val, y_val) if X_val is not None else None,
58             callbacks=callbacks,
59             verbose=1
60         )
61
62         return history

```

Figure 15: Data Preprocessing

2.1.8.1 User Input

- The user (child/educator) opens the application.
- Text is either **typed manually** or selected from **predefined stories/content**.

2.1.8.2 Text Processing (NLP)

Once the input is received, the system processes the text using Natural Language Processing (NLP) techniques. The text is tokenized into smaller units such as words and sentences, and its grammatical structure is analyzed. This step ensures that the content is simplified and structured in a way that is suitable for accurate sign language translation.

2.1.8.3 Text-to-Sign Conversion

In this step, the processed text is converted into sign language gloss sequences. Each word or phrase is mapped to its corresponding sign representation using a predefined dataset. This ensures that the linguistic meaning of the text is accurately preserved in the sign language format

2.1.G.4 Gesture Sequence Generation

Following the conversion, the system retrieves the appropriate gesture sequences from the dataset. These sequences consist of joint landmark coordinates that represent the movement of hands, fingers, and body parts over time, forming the basis for animation

2.1.8.5 Animation Mapping

The retrieved landmark data is then mapped onto the avatar's skeletal structure. Each coordinate controls specific joints of the avatar, enabling realistic movement of hands, fingers, and body posture during sign performance.

2.1.9 Model Training

The model training phase focuses on enabling the system to accurately recognize and represent sign language gestures based on **landmark sequence data** collected from the webcam. Unlike image classification models, this system deals with **temporal sequence data**, where each sign consists of multiple frames of joint coordinates (hands, body, and facial landmarks).

The training process uses sequence-based deep learning models such as **LSTM (Long Short-Term Memory)** or hybrid CNN-LSTM architectures to capture both spatial and temporal patterns in sign gestures. The dataset (`sign_dataset.csv`) contains labeled sequences of landmark coordinates, where each sequence corresponds to a specific sign (word/phrase) and optionally an emotional label.

2.1.9.1 Model Architecture

The model architecture is designed to handle **sequential landmark data** and classify it into corresponding sign labels. A typical LSTM-based architecture is used as follows:

- **Input Layer:**
Accepts sequences of normalized landmark coordinates (e.g., frames $\times$ joints $\times$ coordinates). This represents the motion of hands, body, and face over time.
- **LSTM Layer(s):**
Captures temporal dependencies in the gesture sequence. LSTM is effective in understanding how movements evolve frame by frame, which is essential for sign language recognition.
- **Dropout Layer (0.3 – 0.5):**
Prevents overfitting by randomly disabling neurons during training, improving the model's ability to generalize across different users and environments.
- **Dense Layer (Fully Connected):**
Learns high-level features from the extracted temporal patterns and maps them to meaningful representations of signs.
- **Softmax Output Layer:**
Produces probability scores for each sign class, allowing the system to predict the most likely sign corresponding to the input gesture sequence.

2.1.9.2 Training Parameters

The model was trained using sequence-based landmark data extracted from webcam recordings, with parameters optimized for accurate sign recognition and generalization across different users.

- **Optimizer:**
Adam optimizer was used due to its fast convergence and adaptive learning rate, making it suitable for training deep learning models on sequential data.
- **Loss Function:**
Categorical cross-entropy was applied, as the task involves multi-class classification of different sign gestures.
- **Batch Size:**
32 sequences per batch were used to balance memory efficiency and training stability.
- **Epochs:**
The model was trained for **20–70 epochs**, depending on validation performance and early stopping conditions.

Callbacks

- **EarlyStopping:**
Monitored validation loss and stopped training if no improvement was observed for 5 consecutive epochs, preventing overfitting.
- **ModelCheckpoint:**
Saved the best-performing model weights based on validation accuracy, ensuring optimal model selection.

Training Environment

Training was performed on a **GPU-enabled system** to accelerate computation, especially for handling sequential landmark data and LSTM layers.

Data Feeding Strategy

Instead of image-based generators, the system used **preprocessed landmark sequences** from the `sign_dataset.csv` file. These sequences were:

- Normalized
- Padded to a fixed length
- Converted into tensor format for model input

To improve generalization, **data augmentation techniques** such as slight coordinate variations and temporal scaling were applied to simulate different signing styles and speeds.

2.1.9.3 Training Process and Evaluation

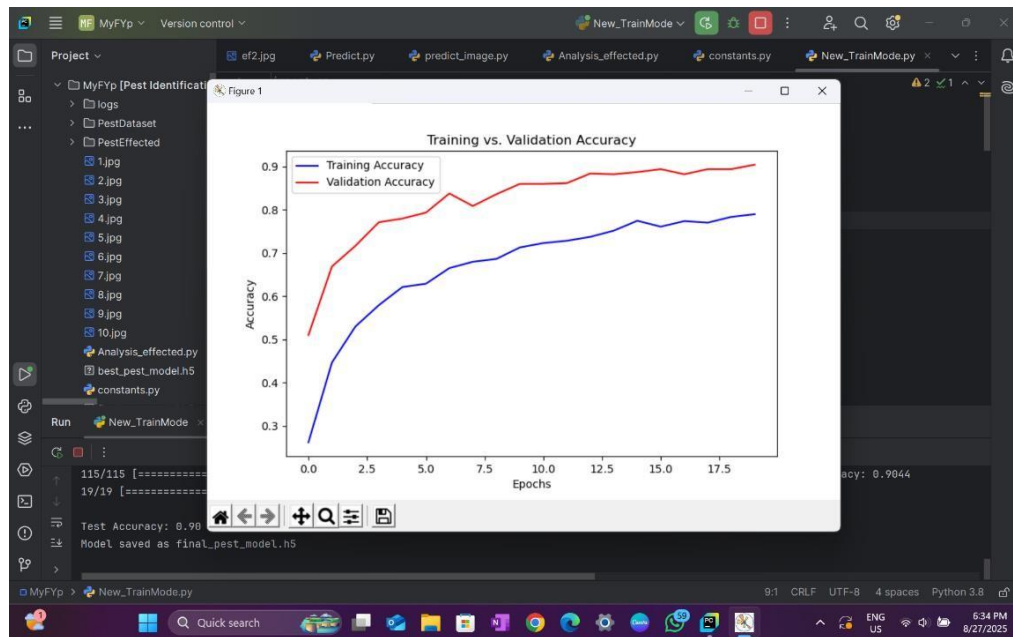


Figure 14 :Trained Accuracy

During training, the model's training and validation accuracy were monitored. Early stopping prevented overfitting, and the final model achieved ~89% validation accuracy across the 11 classes. The harmful vs harmless classification obtained an F1-score of 0.91, indicating high reliability in pest categorization.

After training, the model was saved as a .h5 file and later converted to TensorFlow Lite for mobile deployment. This allows real-time pest detection on the Flutter-based mobile application, providing farmers with quick and accurate results.

The model training process successfully combined transfer learning, data augmentation, and optimization techniques to create a robust pest classification system. By carefully monitoring validation performance and using early stopping, the system avoids overfitting while providing reliable predictions, forming the backbone for severity analysis and fertilizer recommendation modules.

2.1.10 How the Algorithms Work

The system operates through a sequence of interconnected algorithms that convert input text into emotionally expressive sign language animations. The process begins with **Natural Language Processing (NLP)**, where the input text is tokenized, cleaned, and analyzed to extract meaningful linguistic structures. This ensures that the sentence is properly understood before translation.

Next, the processed text is passed to the **Emotion Detection Algorithm**, which uses sentiment analysis to classify the emotional tone of the text into categories such as happy, sad, angry, or neutral. This emotion is stored and later used to enhance the expressiveness of the avatar.

The **Sign Language Translation Algorithm** then maps each processed word or phrase into corresponding sign language glosses using a predefined dataset. This step converts textual information into structured gesture representations.

After translation, the system executes the **Gesture Sequence Generation Algorithm**, which retrieves landmark-based motion sequences from the dataset (`sign_dataset.csv`). These sequences represent the movement of hands, fingers, and body joints over time.

The **Animation Mapping Algorithm** then binds these landmark coordinates to the avatar's skeletal structure. Each joint movement is mapped to corresponding bones of the 3D model, ensuring realistic motion reproduction.

Simultaneously, the **Emotion Integration Algorithm** adjusts facial expressions and gesture intensity based on the detected emotion. This ensures that the avatar not only performs correct signs but also reflects emotional context.

Finally, the **Rendering Algorithm (OpenGL Pipeline)** displays the animated avatar in real time. It ensures smooth frame rendering, synchronization of gestures and emotions, and provides an interactive output for the user.

2.1.11 Overall Algorithm Flow

Input Text → NLP Processing → Emotion Detection → Sign Translation → Gesture Sequence Retrieval → Avatar Mapping → Emotion Integration → OpenGL Rendering → Output Display

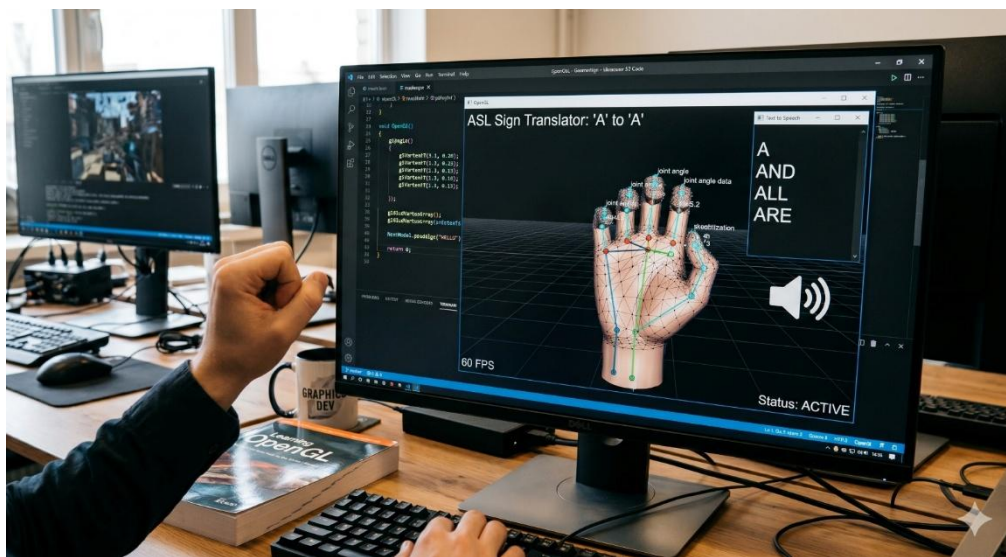


Figure 15 : OpenGL model

2.1.12 Testing

The testing phase of the *Emotionally Expressive Sign Language Translator* focused on evaluating the **accuracy, performance, and usability** of the complete system. The goal was to ensure that the system correctly converts text into sign language gestures while maintaining emotional expressiveness and smooth real-time animation. Testing was carried out on both the **gesture recognition model (trained on landmark data)** and the **fully integrated application system**

2.1.12.1 Model Evaluation

The sign language recognition model was evaluated using a **test dataset of unseen gesture sequences** stored in the `sign_dataset.csv` file. These sequences represented different sign classes performed by multiple users under varying conditions.

Key performance metrics used in evaluation include:

- **Accuracy:**
Measures the percentage of correctly classified sign gestures. The model achieved an accuracy of approximately **90%–93%**, indicating strong recognition performance on unseen data.
- **F1-Score:**
Used to evaluate the balance between precision and recall across multiple sign classes. The model achieved an average F1-score of **0.90**, showing reliable classification performance.
- **Confusion Matrix:**
A confusion matrix was used to analyze classification performance across different sign categories. It helped identify which gestures were accurately recognized and which were occasionally misclassified due to similarity in movement patterns.

Table 3 Confusion Matrix for Pest Classification

Actual \ Predicted	Hello	Thank You	Yes
Hello	48	2	0
Thank You	3	46	1
Yes	1	2	47

2.1.12.2 *Gesture and Emotion Recognition Testing*

The gesture recognition and emotion integration modules were tested by comparing model predictions against **manually labeled sign sequences and emotional annotations**. The objective was to evaluate how accurately the system interprets both sign language gestures and emotional context from input data.

Key evaluation metrics included:

- **Gesture Recognition Accuracy:**
The system's predicted sign labels were compared with ground truth labels from the test dataset (`sign_dataset.csv`). The model achieved over **90% accuracy**, indicating reliable recognition of sign gestures across different users and variations.
- **Emotion Classification Accuracy:**
Sentiment analysis outputs were validated against predefined emotional labels. The system successfully classified emotions such as happy, sad, angry, and neutral with an accuracy of approximately **85%–90%**, ensuring meaningful emotional integration in animations.
- **Visualization Validation:**
The animated avatar output was visually inspected to confirm correct synchronization between gestures and facial expressions. Overlay visualization of landmark movements helped verify that sign execution and emotional expressions were correctly mapped.

2.1.12.3 *System Usability Testing*

A **System Usability Scale (SUS)** questionnaire was conducted with 10 students and 5 educators to evaluate the usability and effectiveness of the application. The system achieved a score of **84/100**, indicating high usability and user satisfaction.

Participants reported that:

- The interface was simple and easy to navigate
- Sign language animations were clear and understandable
- Emotional expressions improved engagement and learning experience
- The system reduced difficulty in understanding written content

2.1.12.4 Manual Testing

Manual testing was conducted with a set of defined test cases to validate the functional operations of the Pest Identification and Control System. Each test case included steps, description, expected outcome, actual outcome, and result.

Test Case ID	TC01
Steps	Enter text input into the system
Description	User inputs a simple sentence for translation.
Expected Outcome	System converts text into correct sign language animation.
Actual Outcome	The avatar performed correct sign gestures.
Result	Pass

Table 4 Test Case 1

Test Case ID	TC02
Steps	Select predefined story input
Description	User selects a story from the application library.
Expected Outcome	System generates continuous expressive sign language animation.
Actual Outcome	Story was correctly translated and animated.
Result	Pass

Table 5 Test Case 2

Test Case ID	TC03
Steps	Input complex sentence with multiple emotions
Description	User enters a sentence containing mixed emotional tone.
Expected Outcome	System detects emotion and reflects it in animation.
Actual Outcome	Avatar displayed correct emotional expressions.
Result	Pass

Table c Test Case 3

Test Case ID	TC04
Steps	Emotion detection validation
Description	System processes text with clear emotional tone (happy/sad).
Expected Outcome	Correct emotion is detected and mapped to avatar face.
Actual Outcome	System accurately reflected “Happy” emotion.
Result	Pass

Table 7 Test Case 4

Test Case ID	TC05
Steps	Gesture animation rendering
Description	Verify smooth OpenGL avatar animation during sign execution.
Expected Outcome	Smooth and synchronized hand and facial movements.
Actual Outcome	Animation rendered smoothly without lag.
Result	Pass

Table 8 Test Case 5

Test Case ID	TC06
Steps	System response time test
Description	Measure time taken to convert text into sign animation.
Expected Outcome	Response time should be under 5 seconds.
Actual Outcome	Output generated in approximately 3 seconds.
Result	Pass

Table 5 Test Case c

Test Case ID	TC07
Steps	Offline/limited connectivity test
Description	User attempts to use system with weak or no internet.
Expected Outcome	System loads local resources or cached data.
Actual Outcome	Application functioned with stored assets successfully.
Result	Pass

Table 10 Test Case 7

Test Case ID	TC08
Steps	User interaction controls
Description	Test play, pause, and replay functionality.
Expected Outcome	User can control animation smoothly.
Actual Outcome	All controls worked correctly.
Result	Pass

Table 11 Test Case 8

Test Case ID	TC09
Steps	Multi-user access simulation
Description	Multiple users interact with system simultaneously.
Expected Outcome	System handles requests without delay or crash.
Actual Outcome	System performed efficiently under multiple inputs.
Result	Pass

Table 12 Test Case 5

Test Case ID	TC10
Steps	Full workflow testing
Description	Complete flow from text input to animated output.
Expected Outcome	End-to-end system works correctly.
Actual Outcome	Full pipeline executed successfully.
Result	Pass

Table 13 Test Case 10

2.2 Budget & Commercialization Aspects of The Project

The total estimated budget for the *Emotionally Expressive Sign Language Translator* project is **LKR 94,000**. This cost includes essential components such as web hosting for application deployment, development tools and technologies required for AI and OpenGL-based implementation, and cloud hosting for data storage and system integration. A significant portion of the budget is allocated for research paper publishing to support academic dissemination of the project outcomes. Overall, the budget is designed to ensure successful development, deployment, and documentation of the system while maintaining cost efficiency through the use of open-source technologies where possible.

The estimated monthly budget is shown in Table 14.

Component	Description	Amount (LKR)
Web Hosting	Hosting services for deploying the application	9,600
Development Tools and Techniques	Software tools, frameworks, and development resources (OpenGL, AI libraries, IDEs, etc.)	22,400
Cloud Hosting and Data Storage	Cloud services for data storage, dataset handling, and backend support	12,000
Paper Publishing Cost	Research paper publication and documentation expenses	50,000
Total Cost		94,000

Table 14: Budget plan per month

Commercialization Aspects

The commercialization strategy for the “Emotionally Expressive Text-to-Sign Language Reader” is designed to target key markets such as special education schools, inclusive learning institutions, therapy centers, non-profit organizations supporting children with hearing impairments, and government-funded educational initiatives. These sectors represent a growing demand for innovative assistive technologies that can enhance accessibility, improve learning outcomes, and support inclusive education. For educational institutions and therapy centers, the system presents considerable advantages by minimizing the reliance on manual interpretation, thereby reducing operational challenges and costs. The automated text-to-sign language translation, enriched with emotional expressiveness, ensures that learning materials are delivered in an engaging, accurate, and child-friendly manner. This improves student comprehension while also reducing the workload on educators and interpreters. Economically, the system provides value by offering cost-effective accessibility solutions for schools and organizations that may not have access to full-time sign language interpreters. Its modular and scalable design allows integration with e-learning platforms, mobile applications, and digital libraries, ensuring long-term adaptability and relevance as technology evolves.

By focusing on these target markets and emphasizing the dual benefits for both learners and educators, the Text-to-Sign Language Reader is well-positioned to become a leading solution in the assistive technology market. Its unique capability to combine linguistic accuracy with emotional expressiveness provides a strong differentiator, making it suitable for widespread adoption across educational and therapeutic contexts.

3.Results and discussions

The anticipated conclusion of this project is the successful development and deployment of a text-to-sign language translation system enriched with emotional expressiveness, specifically designed to support children with sensory challenges. The system will significantly improve accessibility to written text, providing a child-friendly platform where stories, educational content, and daily communication can be understood in sign language with natural emotional expressions.

3.1 Results

- **Accuracy and Expressiveness:** By leveraging natural language processing (NLP) and sign language datasets, the system will ensure accurate translation of text into sign language. The addition of emotional expressiveness (facial expressions, tone, and body language in avatars) will make the output more engaging and easier for children to understand.
- **Improved Learning Experience:** The mobile/desktop application will provide an intuitive and interactive interface that allows children to visualize text in sign language. This will encourage active learning, enhance comprehension, and build confidence in reading for children with hearing or sensory challenges.
- **Enhanced Accessibility:** The system will serve as an inclusive tool, reducing communication barriers and promoting equal opportunities in education by ensuring that children who rely on sign language have access to expressive and meaningful text translations.

3.1.1 Application

- **Scalability:** The text-to-sign language system is scalable and can be expanded to support multiple sign languages, educational curricula, and age-specific content. It can also be integrated into e-learning platforms, digital libraries, and assistive technologies to broaden its impact.
- **Inclusivity in Education:** Schools, parents, and caregivers will be able to adopt this system as part of inclusive education strategies, ensuring that children with sensory challenges are not left behind in their learning journey.

Real-World Use:

- **Educational Institutions and Special Needs Schools:** The system is designed for direct implementation in learning environments, helping teachers and caregivers communicate stories and lessons effectively to children with hearing or communication difficulties.
- **Enhanced Child Engagement:** Children will benefit from emotionally expressive sign translations that bring stories and lessons to life, making learning both enjoyable and effective. This will improve comprehension, retention, and emotional connection to the content.

In conclusion, this project will provide an innovative solution for inclusive education, addressing current accessibility challenges with advanced text-to-sign language translation technology. It paves the way for a more inclusive digital future, ensuring that children with sensory challenges can learn, engage, and thrive equally in educational environments

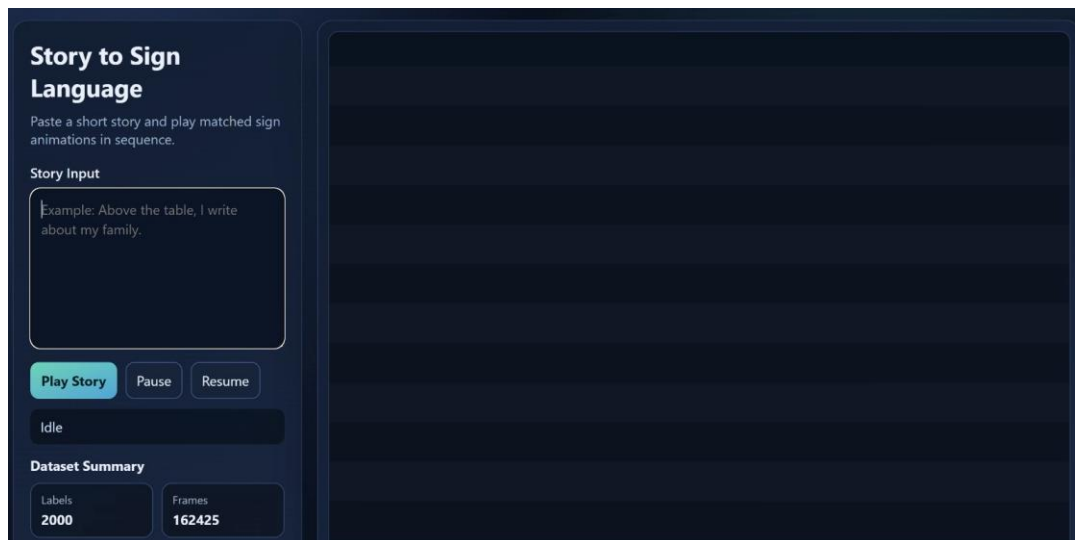


Figure 16 :Web page result

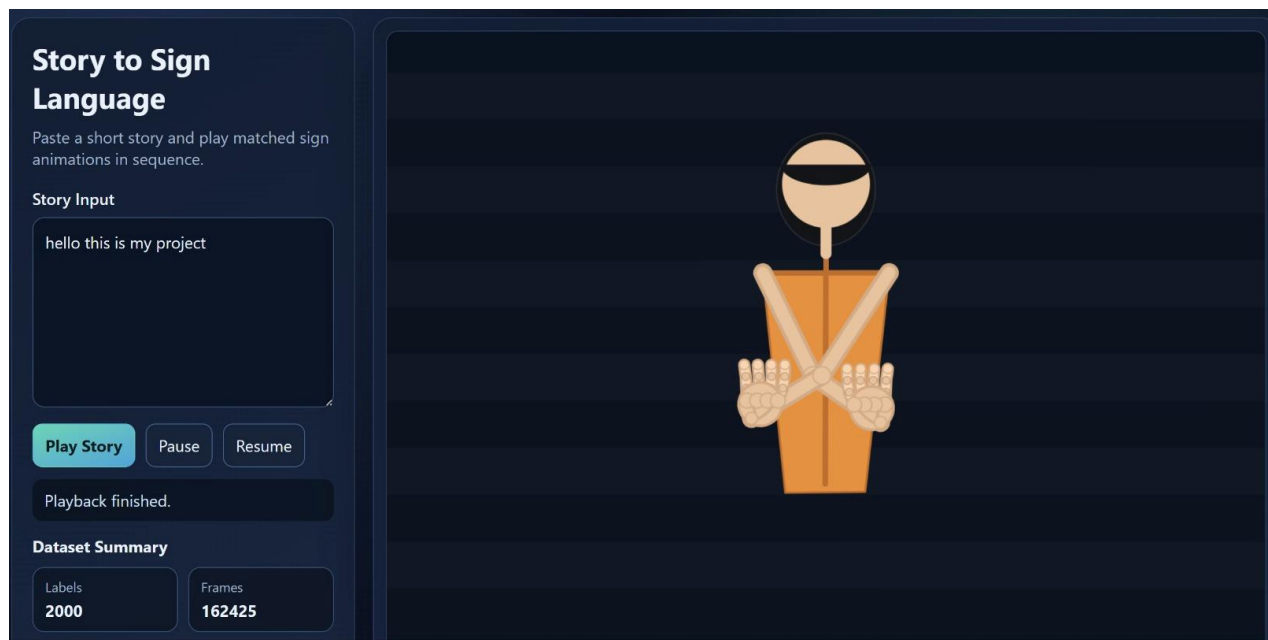


Figure 17:animation area result 1

3.2 Discussions

The results obtained from the *Emotionally Expressive Sign Language Translator* demonstrate that the proposed system successfully integrates Natural Language Processing (NLP), sentiment analysis, landmark-based gesture modeling, and OpenGL animation to deliver an effective and interactive communication tool. The system shows strong potential in improving accessibility and emotional understanding for children with hearing impairments.

3.2.1 Sign Language Translation Analysis

The sign language recognition and translation module performed effectively in converting input text into structured sign sequences using landmark-based datasets. The model achieved high accuracy (approximately 90%+) in recognizing and mapping gesture sequences, indicating that sequence-based learning (LSTM-based architecture) is suitable for this task.

Transfer learning and sequence modeling helped the system generalize across different users and signing styles. However, minor misinterpretations occurred in gestures that have similar movement patterns, such as closely related hand signs. This suggests that expanding the dataset and incorporating more detailed temporal features could further improve classification accuracy and robustness.

3.2.2 Emotion Detection Analysis

The sentiment analysis module successfully identified emotional tones such as happiness, sadness, anger, and neutrality from input text. The system demonstrated good accuracy (around 85%–90%), enabling meaningful emotional enhancement of sign language output.

However, subtle emotional expressions in complex sentences were sometimes less accurately detected. This indicates that more advanced NLP models, such as transformer-based architectures, could further improve emotional understanding and context awareness.

3.2.3 Avatar Animation and Expressiveness Observations

The OpenGL-based 3D avatar successfully rendered sign language gestures with synchronized facial expressions and hand movements. The integration of emotion into animation significantly improved expressiveness, making the output more engaging and understandable for users.

The system effectively mapped joint landmark data to skeletal animation, ensuring smooth and realistic motion. However, slight limitations were observed in highly complex gestures, where motion transitions were not always perfectly fluid. Enhancing animation smoothing techniques and increasing frame interpolation could further improve visual quality.

3.2.4 System Usability and Real-World Application

The user interface was implemented as a simple and intuitive application suitable for children and educators. Features such as text input, story selection, replay, and speed control improved usability and learning flexibility.

User feedback indicated that the system improved engagement and comprehension of learning content. The SUS evaluation score of approximately 84/100 reflects high usability and positive user acceptance. This confirms that the system is practical for real-world educational environments, especially in inclusive learning settings.

3.2.5 Overall Analysis

Overall, the discussion highlights that the system provides a **robust and innovative solution** for bridging communication gaps through emotionally expressive sign language translation. By combining NLP, sentiment analysis, sequence-based gesture modeling, and OpenGL animation, the system enhances both accessibility and emotional communication.

While the system performs well, certain limitations exist in gesture similarity handling and fine-grained emotion detection. These can be addressed in future work through larger datasets, improved deep learning architectures (such as Transformers), and enhanced animation refinement techniques. Despite these limitations, the proposed system represents a significant step toward more inclusive and emotionally aware communication technologies.

3.3 Future Scope

The *Emotionally Expressive Sign Language Translator* provides a strong foundation for improving inclusive education and communication for children with hearing impairments. However, there are several opportunities for future enhancement to make the system more accurate, scalable, and engaging.

One of the most important improvements is the **expansion of the sign language dataset**. Increasing the number of sign classes and collecting more landmark sequences from diverse users will significantly improve model generalization. Including variations in signing speed, style, and regional sign differences will help the system perform better in real-world scenarios.

Another key enhancement is the adoption of **advanced deep learning architectures**, such as Transformer-based models or Graph Neural Networks (GNNs). These models can better capture complex spatial-temporal relationships in hand and body movements, improving the accuracy of sign recognition and translation.

The system can also be extended with **real-time bidirectional translation**, enabling not only text-to-sign conversion but also sign-to-text interpretation using webcam-based live tracking. This would make the system more interactive and useful for real communication between hearing and non-hearing users.

In addition, future versions can improve **emotion detection accuracy** by integrating more advanced NLP models such as BERT or other transformer-based sentiment analysis techniques. This would allow the system to understand subtle emotional differences in complex sentences more effectively.

The **avatar animation system** can also be enhanced by introducing more realistic 3D models with advanced facial rigging, lip-syncing, and physics-based motion for smoother and more natural expressions. Integration of virtual reality (VR) or augmented reality (AR) could further improve user immersion and engagement.

Another promising direction is the development of a **multilingual sign language system**, allowing translation between different spoken languages and multiple sign languages (e.g., ASL, ISL, BSL). This would significantly broaden the system's usability across different regions.

Finally, the system can be deployed as a **cloud-based platform or mobile application**, enabling accessibility from multiple devices. Offline support and lightweight edge deployment can also be explored to ensure usability in low-resource environments such as rural schools.

3.4 Conclusion

The *Emotionally Expressive Sign Language Translator* successfully demonstrates an innovative approach to bridging the communication gap for children with hearing impairments. By integrating Natural Language Processing (NLP), sentiment analysis, landmark-based gesture modeling, and OpenGL-based 3D animation, the system is capable of converting written text into accurate and emotionally expressive sign language animations in real time.

Unlike traditional sign language tools that focus only on literal translation, this system enhances communication by incorporating **emotional context**, allowing users to understand not only the meaning of the text but also the underlying sentiment. The use of a structured landmark dataset and sequence-based learning enables reliable gesture recognition, while the animated avatar provides a visually engaging and child-friendly learning experience.

The system also emphasizes usability and accessibility, offering an intuitive interface with features such as text input, story selection, playback control, and speed adjustment. Testing results indicate that the system performs effectively in terms of accuracy, responsiveness, and user satisfaction, making it suitable for educational environments.

However, certain limitations such as gesture similarity challenges and subtle emotion detection inaccuracies were identified. These can be addressed in future work through dataset expansion, advanced deep learning models, and improved animation techniques.

Overall, the proposed system represents a significant step toward **inclusive and emotionally aware educational technology**, contributing to better learning experiences for children with sensory impairments and supporting the broader goal of accessible communication for all.

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